**Wii-Hab**
A relatively new and evolving field of medicine incorporating interactive video game systems to treat or research neurological or orthopedic ailments

**http://goo.gl/PvaCQ**
Poker Surface uses mobile phones and a multi-touch surface to play interactive card games

# New-age interfaces

## We have been tied to our input devices for far too long, but some cutting edge research in the field of gesture based computing is bringing it out of the realm of science fiction and into our everyday lives, taking us one step closer to our machines

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When someone mentions the futuristic Hollywood film, 'Minority Report', the one visual that sticks out, in that mélange of science fiction, is that of Tom Cruise standing in front of a transparent screen and using hand gestures to manipulate data projected onto that screen. Seeing him do all that without any input device, apart from his two illuminated finger tips, left a lot of us impressed.

Every year millions are spent in research to make input devices more ergonomic. What if there were no input devices? Imagine a world where you do not have to click any button to generate a response, a world where your bodily gestures work as inputs. That is the very idea that has given birth to the genre of computing called Gesture based computing.

Gaming giant Nintendo put it on the popular consumer electronics map with their fifth home gaming console – Wii. It used a wireless controller which mimicked your gestures and translated them into on-screen controls such that while playing tennis, you had to assume that you are holding a racquet instead of a controller and behave accordingly to play the game. This year saw the launch of Sony Playstation Move - a controller based gesture gaming tool and Microsoft Kinect which comes without any controller and has raised the bar for gesture gaming. In this article, we will not talk about gesture gaming as that has got a lot of coverage in the past.

### From Reel to Real

John Underkoffler, the science adviser behind the Minority Report computer interface, showcased a working model of that system at TED in June 2010. His company Oblong has come out with a working model called G-speak that uses a mounted camera which tracks the user's glove as it moves in space. It can handle multiple users and has sub-millimeter precision, allowing users to collaborate on one screen simultaneously. Possible applications of this technology involve analyzing data sets, working on 3D models among other things.

While G-speak still requires you to use gloves to operate the screen, Underkoffler promised a future where you would need no gloves as computer tracking would enable mere hand gestures to do the work.. A researcher at MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL) has improvised on this gesture based technology by hacking into Microsoft's Kinect. The controller-less attribute of Kinect has given birth to a lot of hacks which have been making news in the recent past. MIT engineer Garratt Gallagher used Kinect sensor to differentiate both hands and its fingers in a screen comprising of about 60,000 points. His system uses Kinect alongwith Libfreenet driver for interfacing with Linux. This graphical user interface and hand detection software has been developed at MIT to interface with the open source Robotic Operating System (ROS). ROS is a community-driven open source effort which comprises of a collection of software packages that help robotics research and development. The hand detecting software shows the abilities of the Point Cloud Library (PCL) which is an open-source software package built as a part of ROS. The potential of this research involves controlling real-life robots with gestures thanks to the real-time interaction that can track hand and finger movement at a high speed of 30 frames per second.

**Interesting links around ROS**

**http://goo.gl/r2JoT**
Garratt Galagher's system in action
**http://goo.gl/DiBui**
Everything you wanted to know about ROS and PCL
**http://goo.gl/RSuhU**
A demo of the combined use of ROS and Kinect

### It's all in the glove

The gloves that were seen in G-Speak are special and are fitted with sensors on the tips of the fingers, which help in tracking movement on the screen. MIT undergraduate student Robert Wang along with his associate professor Jovan Popovi  has developed a system of using hand gestures to interact with 3D models using a pair of lycra gloves. Using only a webcam and a pair of lycra gloves with blotches of colour on them, the team has developed a system which can translate gestures made by the gloved hand into corresponding gestures of a 3D model of the hand on screen with little or no lag time.

The glove which has 20 irregularly shaped patches of 10 different colours is a definite fashion faux pas, but it works wonders. The arrangement and shapes of the patches is chosen in such a way that there is no collision of similar coloured patches. An algorithm to rapidly look up visual data in a database is another important component of the whole project.

After the webcam has captured the image of the worn glove, the software developed by Wang erases the background in such a way that you are only able to see the glove superimposed on a white background. This is then reduced to a 40 x 40 pixel size. Here on, to detect the hand movement the system looks up the data

**http://goo.gl/bhGY9**

Robert Wang demos the lycra glove interface seen below